

Art of Physics Education (AoPE)

The Falling Apples

Mechanics

Lesson 1

Joshua Yu

January, 2025

Lesson 1(A): *The Falling Apples*

Vocabulary

- ❖ **Mechanics:** the branch in physics that studies how objects move and behave
- ❖ **Motion:** the change in position over a period of time
- ❖ **Scalar Quantity:** a value that only contains the magnitude of a measurement
- ❖ **Vector Quantity:** a value that contains both magnitude and direction

Scalar Quantities:

- ❖ **Distance** (d , meters): the total length that an object travels
- ❖ **Time** (t , seconds): the duration of an object's motion covering a certain distance
- ❖ **Speed / Rate** (r , meters/second): how fast an object is moving: $r = \Delta d / \Delta t$
- ❖ **Average Speed** ($\bar{r}$): total distance over total time: $\bar{r} = d / t$

Vector Quantities:

- ❖ **Displacement** (Δx , meters): the shortest distance from the final location to the initial location: $\Delta x = x_f - x_i$
- ❖ **Velocity** (v , meters/second): the speed and direction of the motion: $v = \Delta x / t$
- ❖ **Instantaneous Velocity:** the velocity at a particular time

Motion

- ❖ Welcome to Mechanics! Mechanics is all about how objects move and behave.
- ❖ Motion may be loosely defined as changing location throughout a period of time. Motion is taking place whenever something is moving.
- ❖ Things that are static do not move; they stay where they are. They do not have motion.
- ❖ The three facts below are the basis for all mechanics:

Theory	Proof
1. A static object tends to remain static.	If a car starts suddenly, the driver leans back.
2. A moving object tends to keep moving.	If a car stops suddenly, the driver leans forward.
3. A push or pull makes an object start, stop, or change its motion.	Pushing a static swing makes it move. Pulling the moving swing makes it stop. Pushing it harder makes it move faster.

- ❖ The first point is obvious. Things won't move unless they are pushed or pulled.
- ❖ The second point also makes sense. This is why seatbelts are necessary on cars. If the car suddenly stops, the seatbelts prevent the driver and others in the vehicle from leaning forward which may cause injury. However, moving objects eventually slow down. This doesn't disprove the second theory because there are other pushes or pulls acting on the object...
- ❖ The third point explains itself. Pushing harder causes an object to move or change its motion. Pushing from the left makes the object move to the right.

Common Mathematical Notations

- ❖ x_0 : the starting measurement of x . Usually $x_0 = 0$. Pronounced "x naught."
- ❖ x_i : the initial measurement of x . Pronounced "x initial."
 - Note that x_i can refer to any initial point in a specific range. It does not have to be the very beginning point. The beginning point is denoted as x_0 . However, these two can be used interchangeably.
- ❖ x_f : the final measurement of x . Pronounced "x final".
- ❖ Δx : change in x : $\Delta x = x_f - x_i$ or $\Delta x = x_f$ if $x_i = x_0$.
- ❖ $\bar{x}$: the average value of x .

Scalar and Vector Quantities

- ❖ A scalar quantity is a value that only contains magnitude. Magnitude refers to how much or how many of whatever is being measured. Scalar quantities can only be positive or 0.
- ❖ A vector quantity not only includes magnitude, but also direction. Direction refers to the way or heading the object is moving in. For example, a positive vector represents moving forward, but a negative vector represents moving backward. Vectors can be positive, negative, or 0.

Scalar Variables

- ❖ Distance is the total length that an object travels. It is a scalar quantity, represented by the variable d with the SI unit in meters (m). Clearly, $d \geq 0$ because negative distance does not exist. So $d = |x_f - x_i|$.
- ❖ Time is the duration of the object's motion covering a certain distance. It is a scalar quantity, represented by t with the SI unit in seconds (s). Clearly, $t \geq 0$ because negative time does not exist.
- ❖ Speed or rate denotes how fast an object is moving. It is a scalar quantity, represented by r with the SI unit in meters/second (m/s). The mathematical definition of speed is: $r = \Delta d / \Delta t$. The speed over a particular distance during a particular time frame can be calculated, so Δ symbols are used. Since both d and t are positive, $r \geq 0$.
- ❖ Alternatively, the average speed ($\bar{r}$) equals the total distance over total time: $\bar{r} = d/t$.

Vector Variables

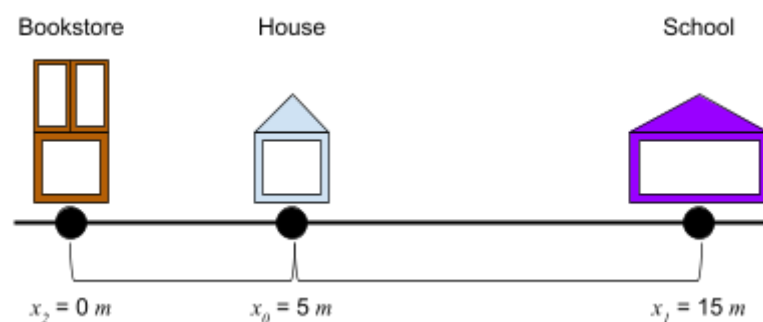
- ❖ Vector variables are primarily used in Mechanics because it also includes direction.
- ❖ Displacement is the shortest length between the final position (x_f) and the starting position (x_0) of the entire journey of an object, so it is denoted as Δx . Its SI unit is still meters (m). Since displacement is a vector quantity, it shows in which direction the object ends up relative to its initial location and the positive direction. Thus it is very important to define which direction is the positive direction. For linear motion, defining one direction to be positive and the opposite as negative is enough. For example, let East be positive. If the object ends up to the east of its original location, the displacement is positive. If it is to the West, the value is negative.
 - A positive displacement does not imply the object is constantly moving east. It may move west and then move east past its original location. Displacement only deals with the final location compared to the initial location.

- ❖ Velocity refers to the speed and the direction of which the object is moving. It is denoted as v with the SI unit in meters/second (m/s). Unlike speed, velocity tells the direction that the object is moving over a specific distance and time. A positive velocity means that the object is moving in the positive direction. A negative velocity means that the object is moving in the negative direction. Velocity is the displacement over a certain time frame:

$$v = \Delta x / \Delta t.$$
- ❖ The instantaneous velocity is the velocity at a particular time.

Example Problems

Joshua walks from the house to the school and then to the bookstore. The number line with the values are shown at right.



What is the distance traveled?

We can treat this as a 2-step problem.

→ From the house to the school: $d_1 = |x_1 - x_0| \Rightarrow |15 - 5| = 10$

→ From the school to the bookstore: $d_2 = |x_2 - x_1| \Rightarrow |15 - 0| = 15$

Total distance: $d = d_1 + d_2 = 10 + 15 = 25 \text{ m}.$

What is the displacement?

We only need to consider the original and the final locations.

We will define moving to the right as the positive direction. Since the bookstore is to the left of the house, we expect to get a negative answer.

Displacement: $\Delta x = x_f - x_0 = x_2 - x_0 = 0 - 5 = -5 \text{ m}.$ The bookstore is 5 m to the left of the house. If moving to the left is the positive direction, the displacement will be + 5 m.

Joshua takes 30 seconds to walk from the bookstore to the school. Find the speed.

Distance from the school to the bookstore: $d = |x_1 - x_2| = |15 - 0| = 15 \text{ m}.$

Time: $t = 30 \text{ s}$

Speed: $r = d/t = 15/30 = 0.5 \text{ m/s}.$ Notice that the Δ symbols are unnecessary since the initial measurements are 0.

Joshua takes 3 seconds to run from the school to the bookstore. Find the velocity.

We will define moving left as the positive direction here. In this case, $x_1 = x_0 = 0$ and

$$x_f = x_2 = +15.$$

$$\text{Displacement: } \Delta x = x_2 - x_1 = 15 - 0 = +15 \text{ m.}$$

$$\text{Time: } t = 3 \text{ s.}$$

Velocity: $v = \Delta x/t = 15/3 = +5 \text{ m/s}$. This shows that his speed is 3 m/s and that he is moving west. The positive velocity value represents moving in the positive direction.

Find the average speed if it takes him 8 seconds to go from the house to the school and 12 seconds from the school to the bookstore.

Distance from the house to the school: 10 m . Time: 8 s .

Distance from the school to the bookstore: 15 m . Time: 12 s .

Total distance: $10 + 15 = 25 \text{ m}$. Total time: $8 + 12 = 20 \text{ s}$.

$$\text{Average speed: } \bar{r} = d/t = 25/20 = 1.25 \text{ m/s.}$$

Notice that the average speed is not the average of the two speeds calculated separately. This is because the time it takes for the two journeys is unequal. The average speed is the average of the two speeds only if the time in both instances is the same.

Problem-Solving Procedures

- ❖ Correctly establish the initial/starting and final quantities when using the Δ symbol.
- ❖ Always define which direction is positive. In linear mechanics, one positive direction is enough.
 - Any direction can be defined to be positive. The opposite direction will be negative.
 - Do not change the definition of the positive direction. Define it and stick with it through the entire calculation.
- ❖ Be clear and understand the definitions of vector variables.
- ❖ Convert all units to SI units before calculation if necessary.

Lesson 1(B): *The Falling Apples*

Vocabulary

- ❖ **Force**¹ (F , Newtons, vector): a push or pull
 - **Agent**: the object which creates or exerts the force
 - **Victim**: the object which receives the force
 - **Net Force**: the total amount of force acting on an object
- ❖ **Equilibrium / Balanced Force**: the condition when the net force is 0
 - **Static Equilibrium**: the equilibrium when the object is static
 - **Dynamic Equilibrium**: the equilibrium when the object is moving
- ❖ **Gravity**: a force which makes objects fall vertically down to the ground
- ❖ **Free Fall**: motion when the only force acting on the object is gravity (air resistance is negligible)
- ❖ **Resistance**: a force which opposes motion (such as friction or air resistance)
- ❖ **Friction**: a resistance force created when one object moves in contact with another

- ❖ **Mass** (m , kilograms, scalar): the amount of matter or inertia an object has
- ❖ **Inertia**: the tendency or inherent laziness of an object to keep its original motion
- ❖ **Newton's 1st Law of Motion (Law of Inertia)**: "An object must remain stationary or move in a straight line at a constant speed unless it is subject to an unbalanced force"

- ❖ **Acceleration** (a , meters/second², vector): change in velocity over time: $a = \Delta v/t$

Forces

- ❖ A force is a push or pull. It is a vector quantity. Anything that moves is mostly likely due to one or more forces acting on objects. Force is denoted as F and has the unit of Newtons (N).
- ❖ A force can make an object start moving. A force in the same direction as the object's motion makes the object move faster.
- ❖ Every force must have an agent and a victim. The agent creates or exerts the force. The victim receives the force.

¹ Forces will be discussed in much more detail in Lesson 2 and 3.

- ❖ The vector sum of all forces acting on an object is called the net force. The forces are vectors which may be positive or negative. The net force is the sum of positive and negative forces.
- ❖ When the net force is 0 or completely cancels out, the object is in equilibrium; the forces are balanced. Static equilibrium is the equilibrium when the object is static. Dynamic equilibrium is the equilibrium when the object is moving.
- ❖ If a static object is in equilibrium, it will not move. If a moving object is in equilibrium, it will keep its original velocity and motion.

Resistance

- ❖ Resistance is a force which opposes motion. When a force is applied in the opposite direction of an object's motion, the object will slow down.
- ❖ Friction is the most common resistance. Friction prevents an object's motion when the object contacts another object when it moves.
- ❖ Air resistance can also prevent motion when a light object falls to the ground. Air particles can reduce an object's moving speed.

Mass and Inertia

- ❖ Mass is the measure of how much matter an object has. It is a scalar quantity, denoted as m , and is measured in kilograms (kg). The more mass, the heavier the object.
- ❖ The mass of the same object is always constant everywhere, but weight is the heaviness of an object due to gravity. The weight can change, but the mass never changes.
- ❖ All objects have inertia, which is the object's intrinsic tendency or inherent laziness to remain in its original motion. Inertia is not affected by gravity, friction, or other forces. Inertia is why static objects want to remain static, and moving objects want to keep moving. Inertia is also not a force.
- ❖ Mass is proportional to inertia. In fact, mass can also be defined as the measure of inertia.
- ❖ Given the same force, the less mass, the faster the velocity change.
- ❖ Given the same mass, the more force, the faster the velocity change.

Newton's 1st Law of Motion (Law of Inertia)

- ❖ Newton's 1st Law of Motion states: "An object must remain stationary or move in a straight line at a constant speed unless it is subject to an unbalanced force."
- ❖ Mathematically, an object's motion won't change unless the net force is not 0.

Gravity and Falling Objects

- ❖ A static object tends to remain static, but it will move when it loses support. An object falls vertically toward the ground when it loses support.
- ❖ Gravity is the force which makes objects fall vertically down to the ground. All objects that lose support or are falling eventually reach the ground.
- ❖ Free fall is the motion where the only force acting on the object is gravity. During free fall, air resistance is negligible.
- ❖ However, horizontal forces such as friction may also occur. If the support is rough or is removed slowly, the falling object may be shifted in the horizontal direction as well.
- ❖ Because of inertia, lighter objects are more sensitive to friction, whereas heavier objects are not influenced by friction and fall directly vertically.

Summary: Motion

- ❖ An object can move because of its initial velocity, inertia, and forces.
- ❖ An object starts moving if it is subject to a propelling force creating unbalanced forces.
- ❖ An object keeps its motion because of inertia and if it is in dynamic equilibrium.
- ❖ An object stops moving if it is subject to a resisting force creating unbalanced forces.
- ❖ An object can change its motion because of unbalanced forces.

Acceleration

- ❖ Objects do not always move at a constant velocity. Acceleration is denoted as a and is defined as the change in velocity over time, or $a = \Delta v/t$. It is a vector quantity and has the SI unit in meters per second squared (m/s^2).
- ❖ A higher acceleration means that the object is speeding up within a shorter time frame.
- ❖ If a car speeds up or slows down suddenly, then the driver will be dangerously affected. He will lean backward and forward when the car speeds up and slows down respectively.

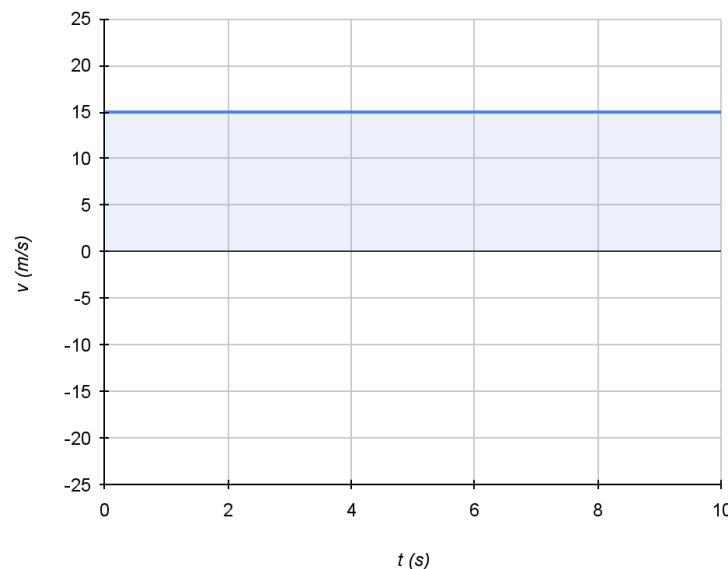
- ❖ Trains do not need seatbelts, because they speed up or slow down gradually over a long period of time. Passengers will not be able to feel sudden movement.
- ❖ We are moving with the earth at about 460 m/s . Even though the speed is very high, we do not feel the motion because there is little acceleration. We do not feel that the earth is moving. When we jump and land back on the earth, our location does not change. Everything is moving together with the earth. There is no force parallel to the earth's surface, so we do not land hundreds of meters behind!

Constant Velocities

- ❖ The definition of velocity is $v = \Delta x/t = (x_f - x_0)/t$. Rearranging gives $\Delta x = vt$ and $x_f = x_0 + vt$.
- ❖ A positive and negative velocity means that the object is moving in the positive and negative directions respectively.
- ❖ The relationship between time and velocity may be graphed.
- ❖ The x-axis is time and the y-axis is velocity.
- ❖ If the velocity is constant, the graph is a straight line.
- ❖ The displacement is the area between the graph and the axes because the displacement is the product of time and velocity.
- ❖ If the velocity change is linear (constant acceleration), then the slope of the line is the acceleration.

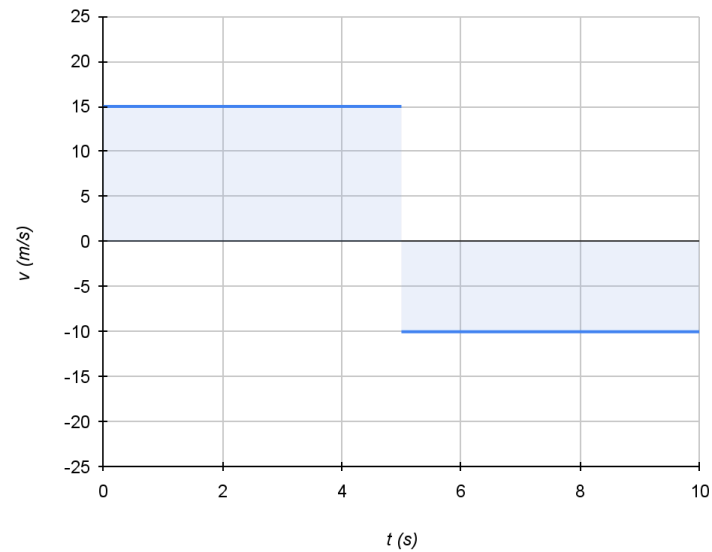
Example 1: Constant Positive Velocity

- ❖ An object travels at constant velocity of positive 15 m/s for 10 s .
- ❖ The displacement is $\Delta x = vt = 150 \text{ m}$. If east is positive, the object is 150 m to the east compared to its original position 10 s later.
- ❖ The slope of the line is 0, so there is no acceleration.



Example 2: Constant Positive and Negative Velocities

- ❖ An object travels at constant velocity of positive 15 m/s for 5 seconds. It suddenly flips direction and travels at a constant velocity of -10 m/s for 5 s.
- ❖ The displacement is the sum of the two areas:
 $\Delta x = (15 \times 5) + (-10 \times 5) = 25 \text{ m}$.
- ❖ The slopes of the lines are 0, so there is no acceleration.



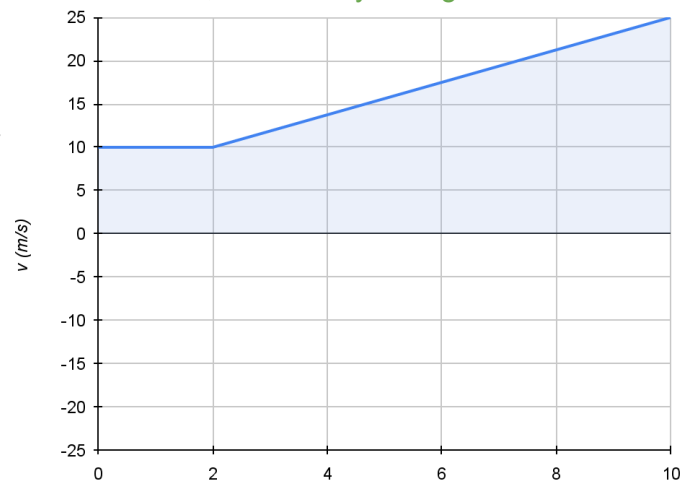
Lesson 1(C): *The Falling Apples*

Vocabulary

- ❖ **Displacement Formula:** $\Delta x = v_0 t + \frac{1}{2}at^2$
- ❖ **Newton's 2nd Law of Motion:** "The net force applied to an object with a fixed mass produces an acceleration proportional to the net force and inversely proportional to the mass." $F_{net} = ma$

Constant Accelerations

- ❖ Acceleration is $a = \Delta v/t = (v_f - v_0)/t$. Rearranging gives $\Delta v = at$. In other words, at is the *velocity change*. If an object travels in constant acceleration, the velocity changes at the same rate.
- ❖ For example, an object is traveling at 10 m/s for 2 s and changes the velocity at a constant rate for 8 s until it reaches 25 m/s.
- ❖ The acceleration is constant throughout the entire velocity change.
- ❖ The acceleration here is $a = (25 - 10) / 8 = (15/8) \text{ m/s}^2$. It is also the slope of the line because the acceleration is constant.



Instantaneous or Final Velocities

- ❖ When the acceleration is known, the instantaneous or final velocity can be found using the equation $\Delta v = at$. Rearranging gives $v_f = v_0 + at$. v_f can be solved if the initial velocity, acceleration, and the change in time are known.
- ❖ For example, the instantaneous velocity v after 2 seconds after the velocity change is:

$$v = v_i + at = 10 + (15/8) \times 2 = (55/4) \text{ m/s}.$$

Negative Accelerations

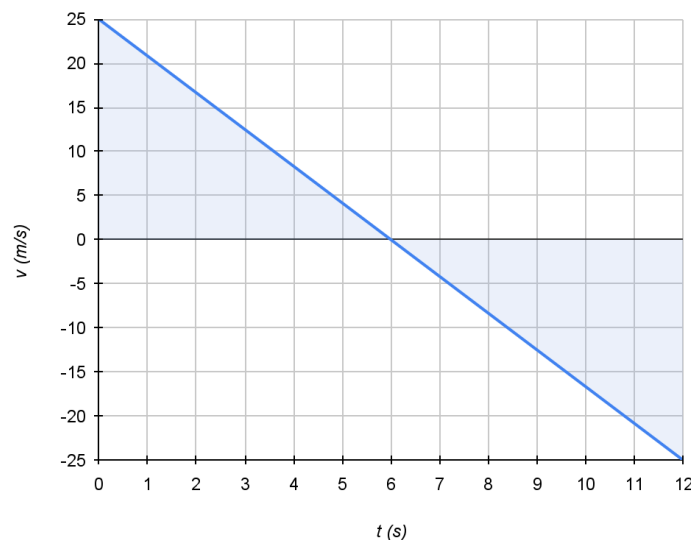
- ❖ It is very important to truly understand the difference between positive and negative accelerations. Acceleration equals the *change* in velocity over time. If $v_f > v_i$, $a > 0$. Similarly, if $v_f < v_i$, $a < 0$.
- ❖ It is incorrect to conclude that a positive acceleration always means the speed increases, while negative acceleration means the speed decreases. What matters is the velocity change.
- ❖ If both the velocity and acceleration are in the same direction (pointing toward the positive or negative direction), the speed increases. If in opposite directions, the speed decreases.
- ❖ Remember that positive velocity means moving in the positive direction. Negative velocity means moving in the negative direction. Thus here are 4 cases and the effect on the speed:

1. If $v_i > 0$ and $a > 0$, they are in the *same* direction. So $v_f > v_i$ and the speed increases.
2. If $v_i > 0$ and $a < 0$, they are in *opposite* directions. So $v_f < v_i$ and the speed decreases.
3. If $v_i < 0$ and $a < 0$, both are in the *same* direction. So $v_f < v_i$, and the speed increases.
4. If $v_i < 0$ and $a > 0$, they are in *opposite* directions. So $v_f > v_i$, and the speed decreases.

- ❖ In this example, an object starts at $v_0 = 25 \text{ m/s}$. After 6 seconds, the velocity becomes 0 m/s . After the following 6 seconds, $v_f = -25 \text{ m/s}$.

- ❖ We see that the plot is a straight line, so the acceleration is constant. The acceleration is $a = \Delta v / t = ((-25) - 25) / 12$
 $= (-25/6) \text{ m/s}^2$.

- ❖ The acceleration is negative. In the first 6 seconds, the object goes from 25 m/s to 0 m/s . The initial velocity is positive but the acceleration is negative. Thus, the speed decreases since $0 \text{ m/s} < 25 \text{ m/s}$.



- ❖ But the velocity goes from $v_i = 0 \text{ m/s}$ to $v_f = (-25) \text{ m/s}$. Here, $v_f < v_i$, so the speed increases.
- ❖ So, the object slows down from 25 m/s to 0 m/s after 6 seconds and travels in the opposite direction while speeding up to -25 m/s .

Calculating Displacement

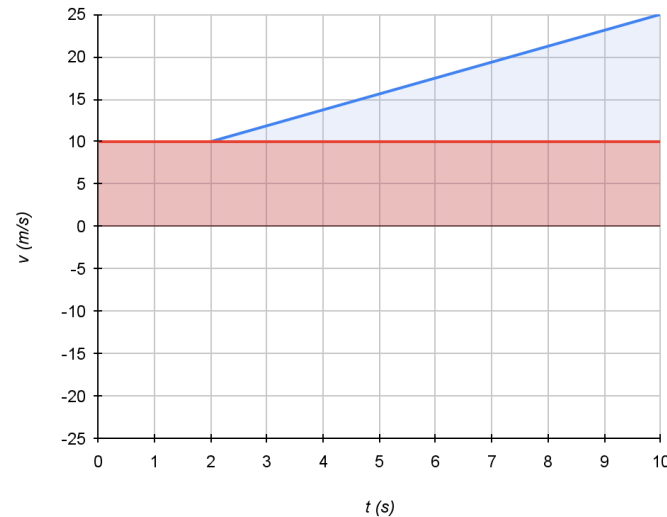
- ❖ Displacement can be considered as the area below the plot and the time axis.
- ❖ To calculate the displacement, break this region into a rectangle (red region) and a triangle (blue region). Thus,

$$\Delta x = (10 \times 10) + \frac{1}{2} (8 \times 15) = 160 \text{ m}.$$

- ❖ Notice that $\Delta v = at = 8 \text{ m/s}$.
- ❖ So we can generalize this equation as the displacement formula:

$$\Delta x = v_0 t + \frac{1}{2} (at)t = v_0 t + \frac{1}{2} at^2$$

- ❖ This equation relates displacement, acceleration, time, and initial velocity together. Obviously, $\Delta x = \frac{1}{2} at^2$ if $v_i = 0 \text{ m/s}$.



Average Velocity During Constant Acceleration

- ❖ The average velocity is the total displacement divided by total time, $\bar{v} = (v_0 t + \frac{1}{2} at^2)/t$
 $= v_0 + \frac{1}{2} at$.
- ❖ From $v_f = v_0 + at$, $\frac{1}{2} at = \frac{v_f - v_0}{2}$, so $\bar{v} = \frac{2v_0}{2} + \frac{v_f - v_0}{2} = \frac{v_0 + v_f}{2}$.
- ❖ Therefore, the average velocity during a constant acceleration is simply $\bar{v} = \frac{v_0 + v_f}{2}$.

Real-World Example: Displacement and Constant Acceleration

A car travels at 28 m/s and brakes at a constant acceleration whose magnitude is 6.6 m/s^2 . What is the displacement that the car travels from the beginning of the velocity change to when the car stops?

We define *positive* as moving forward. Since the car is moving forward, $v_0 = 28\text{ m/s}$ and the car eventually stops, so $v_f = 0\text{ m/s}$. This means $v_f < v_0$, so $a = -6.6\text{ m/s}^2$.

To solve for t , we use $v_f = v_0 + at$, so $t = 4.24\text{ s}$. Finally, we can find the displacement using the equation above: $\Delta x = 59.4\text{ m}$.

Newton's Second Law of Motion

- ❖ Given the same object with the same mass, more force leads to a larger acceleration.
- ❖ Given the same net force, more mass leads to a smaller acceleration.
- ❖ So, the net force is proportional to mass and to acceleration, while mass and acceleration are inversely proportional.
- ❖ This is Newton's 2nd Law of Motion: "The net force applied to an object with a fixed mass produces an acceleration proportional to the net force and inversely proportional to the mass." Mathematically, $F_{\text{net}} = ma$. F is the net force (vector sum) in a particular axis.
- ❖ From this equation, $N = kg \times \frac{m}{s^2}$. This is the definition of a Newton, the unit of force.
- ❖ It takes more force to slow down or speed up a heavier object.
- ❖ Light objects can accelerate to a greater extent when applied the same force.

Falling Objects

- ❖ Two objects with different masses fall at the same acceleration to the ground at the exact same time. For example, two different rocks fall to the ground at the same time.
- ❖ However, hollow or extremely light objects such as paper or feathers take much longer to fall to the ground because of air resistance. Air resistance can cause a light object to fall slower.
- ❖ In most cases, air resistance acting on concrete, solid objects is negligible. So these objects fall to the ground at the same time.

Lesson 1(D): *The Falling Apples*

Vocabulary

- ❖ **Gravitational Acceleration:** $9.8m/s^2$

Kinematic Equations for Constant Acceleration Motions

- ❖ There are 5 extremely useful equations when the acceleration is constant.
- ❖ These equations relate these 4 variables together: Δx , v_0 , v_f , a , t .
- ❖ Each equation relates 4 out of the 5 variables together. Each variable is present in 4 of the 5 equations. All 5 variables may be solved using one or more of these equations if at least 3 variables out of the 5 are known in a problem.

- ❖ Equation 1: $\Delta x = v_0 t + \frac{1}{2} a t^2$. This is the sum of the red rectangular area and the blue triangular area. In this example, $\Delta x = 100 + \frac{1}{2} (2)(10)^2 = 200 \text{ m}$.

- ❖ Equation 2: $\Delta x = v_f t - \frac{1}{2} a t^2$. This is the entire colored area minus the green triangular area. In this example, $\Delta x = 300 - \frac{1}{2} (2)(10)^2 = 200 \text{ m}$.

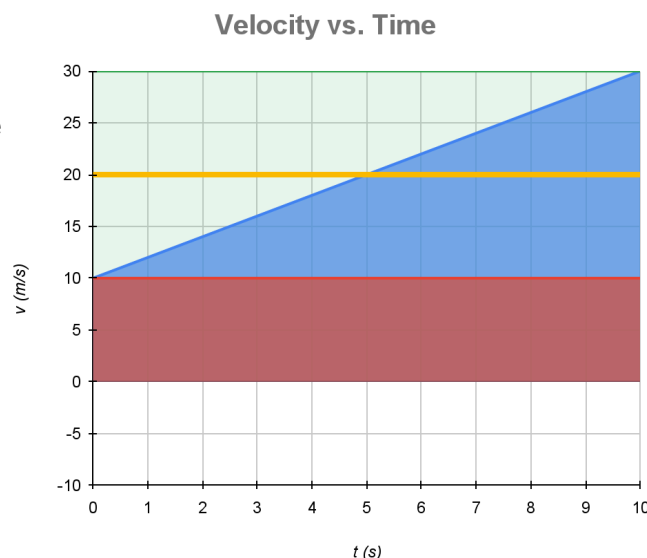
- ❖ Equation 3: $\Delta x = \bar{v} t = \frac{v_0 + v_f}{2} t$. This is the colored area under the yellow line. The yellow line is the average velocity. In this example, $\Delta x = \frac{10+30}{2} \times 10 = 200 \text{ m}$.

- ❖ Equation 4: $v_f = v_0 + at$. This is simply the definition of acceleration. In this example, $v_f = 10 + (2)(10) = 30$

- ❖ Equation 5: $2a\Delta x = v_f^2 - v_0^2$. This equation is derived from Equation 4, $t = \frac{v_f - v_0}{a}$.

Plugging this into Equation 3, $\Delta x = \frac{v_0 + v_f}{2} \times \frac{v_f - v_0}{a}$. Rearranging gives the desired

equation. In this example solving for Δx , $\Delta x = \frac{(30)^2 - (10)^2}{2(2)} = 200 \text{ m}$.



Summary: Kinematic Equations for Constant Acceleration Motions

Each variable only appears in 4 equations. (It is important to memorize them.)

- | | |
|---|--------------------------|
| 1. $\Delta x = v_0 t + \frac{1}{2} a t^2$ | Doesn't need: v_f |
| 2. $\Delta x = v_f t - \frac{1}{2} a t^2$ | Doesn't need: v_0 |
| 3. $\Delta x = \bar{v} t = \frac{v_0 + v_f}{2} t$ | Doesn't need: a |
| 4. $v_f = v_0 + a t$ | Doesn't need: Δx |
| 5. $2a\Delta x = v_f^2 - v_0^2$ | Doesn't need: t |

Problem-Solving Strategies of Constant Acceleration Motions

- Identify the positive x direction.
 - Adhere to the positive direction throughout the entirety of the problem.
- Identify the initial and final points of the problem.
 - It is helpful to set the initial position as $x = 0$.
- Identify the given and non-given variables (Δx , a , t , v_0 , v_f).
 - Convert all measurements into SI units.
 - Check if the problems give 3 known variables.
- Choose which out of the 5 equations to use based on which variable is unknown.
 - For example, if a , t , and v_0 are known, but v_f and Δx are unknown, solve for Δx using Equation 1. Solve for v_f using Equation 4.

Gravity Acceleration for Falling Objects

- ❖ When an object is falling, gravity is the most important force at work. The gravitational acceleration (g) is about 9.8 m/s^2 . Using Newton's Second Law, or $F_{\text{net}} = ma$, the gravity force is the mass of the object multiplied by the gravitational acceleration 9.8 m/s^2 .
- ❖ For compact and solid objects, air resistance is negligible. When falling, $a = g = 9.8 \text{ m/s}^2$.

- ❖ For light, hollow, and objects with large surface area, air resistance cannot be ignored. If downward is the positive direction, $F_{net} = F_{gravity} - F_{air}$. $a \neq 9.8 \text{ m/s}^2$ but is now

$$a = F_{net}/m.$$

- ❖ Therefore, although two compact solid objects with different masses are dropped from the same time and height, they both fall to the ground at the same time, because the gravity acceleration both objects experience is equivalent.

Calculating Gravity Forces

An object on Earth has a mass of 2.5 kg . Find the gravity force exerted on the object.

$$F_{gravity} = mg = (2.5)(9.8) = 24.5 \frac{\text{kg m}}{\text{s}^2} = 24.5 \text{ N}.$$

Example of Constant Acceleration Motion

Toby, standing on the edge of a 50 m cliff, releases a pebble. After a short while, the pebble hits the ground. Find all 5 variables.

Let downward be the positive direction. The initial point is the point where the pebble is released, and the final point is where the pebble hits the ground.

Δx : The ground is 50 m below the cliff. So $\Delta x = 50 \text{ m}$. (Downward is the positive direction.)

a : since this is a pebble, air resistance is ignored. $a = g = 9.8 \text{ m/s}^2$.

v_0 : the pebble was not moving at the starting point. $v_0 = 0 \text{ m/s}$.

t : Using Equation 1 to solve for t , we find that $t \approx 3.19 \text{ s}$.

v_f : Using Equation 4, $v_f = 31.26 \text{ m/s}$.